



BITS Pilani
DIGITAL

READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 13

Data Science and Financial Modelling: Applications and Cases (Derivatives)



STUDY NOTES • EDITION 1.0 • WEEK 13

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How to use these notes



These notes cover the whole of Week 13: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A shows why derivative payoffs defeat straight lines, and introduces the trees and forests that handle them. Part B covers volatility — the one input that cannot be observed — and the simulation used when no formula exists. Part C moves to the desk itself: daily settlement, the Greeks, and the execution problem. This is the final module of the course.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

Contents



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CONSOLIDATION

Module summary, by learning objective

Glossary and self-check

Module overview



Week 11 reviewed equity decisions monthly; Week 12 updated bond valuations nightly. In derivatives a position can be correct in the morning and incorrect by lunchtime. Two things drive that difference: payoffs that bend sharply, which no straight line can represent, and leverage with daily cash settlement, which turns a slow reaction into a forced exit.

One thread runs through all nine topics: **a derivatives position decays even when nothing is traded**. Beta drifts, the basis narrows, volatility shifts and the Greeks move continuously — which is why the tools of this module measure change rather than state.

A note on coverage: the lecture material introduces this week as also covering how to judge honestly whether a model works, and Week 11 refers forward to a detailed treatment of overfitting here. No recorded video covers either. The theme appears throughout — in the greedy tree of Topic 3, in feature importance, and in the simulator caution of Topic 9 — but it is not treated as a separate topic.

Learning objectives

By the end of this module, you will be able to:

- 1 **Explain** why derivative payoffs defeat linear models, and describe decision trees and random forests.
- 2 **Explain** why volatility cannot be observed, and interpret implied volatility and the volatility smile.
- 3 **Describe** Monte Carlo simulation and design a stress test.
- 4 **Explain** daily settlement, the Greeks, and why execution suits reinforcement learning.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	The hedge that quietly stopped working	A · The problem	L41 V1	LO 1
2	Decision trees	A · The tools	L41 V2	LO 1
3	From one tree to a forest	A · The tools	L41 V2	LO 1
4	Volatility: the input you cannot observe	B · Volatility	L41 V3	LO 2
5	The volatility smile and surface	B · Volatility	L41 V3	LO 2
6	Simulation and stress testing	B · Simulation	L41 V4	LO 3
7	The ledger that never sleeps	C · The desk	L42 V1	LO 4
8	The hedge that watches itself	C · The desk	L42 V2	LO 4

Topic	What it covers	Part	Source	Serves
9	Reinforcement learning and execution	C • The desk	L42 V3	LO 4

Key terms at a glance



Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Decision tree

A model learning a sequence of conditional questions, applying different rules in different situations.

Random forest

Many trees built on random samples, their predictions averaged for stability.

Feature importance

A ranking of inputs by how much they contributed to predictions.

Volatility clustering

The tendency of calm to follow calm and turbulence to follow turbulence.

Implied volatility

The volatility forecast hidden inside an option's market price.

Volatility smile

The curved pattern of implied volatility across strikes, higher on the low-strike side.

Fat tails

Extreme market movements occurring more often than a bell-shaped distribution predicts.

Monte Carlo simulation

Generating thousands of possible futures and averaging the resulting payoffs.

Delta

The change in an option's value for a ₹1 move in the underlying.

Vega

The change in an option's value when volatility changes.

Market impact

The adverse price movement a large order causes before it completes.

Execution policy

A learned sequence of trading decisions, rather than a single forecast.

Interaction

Where one factor matters only when another condition also holds.

Gradient boosting

Trees built sequentially, each correcting the previous ones' mistakes.

India VIX

The NSE's real-time measure of expected 30-day volatility, derived from Nifty 50 option prices.

GARCH

A family of models treating volatility as changing over time, fitted by maximum likelihood.

Black-Scholes

The 1973 option pricing model, used in practice in reverse as a translator from price to volatility.

Volatility surface

The three-dimensional structure formed by smiles across several expiry dates.

Volatility crush

The sharp fall in implied volatility once an event resolves the uncertainty.

Stress testing

Simulating deliberately severe and unfavourable conditions.

Gamma

The rate at which delta itself changes.

Theta

The value an option loses as one more day passes.

Market depth

How much can be traded without causing a significant price movement.

PART A • WHY DERIVATIVES NEED DIFFERENT TOOLS

When a straight line will not do

1 The Hedge That Quietly Stopped Working

PART A • WHY DERIVATIVES NEED DIFFERENT TOOLS • TOPIC 1 OF 9 • SERVES LEARNING OBJECTIVE 1



IN ONE LINE

Meera traded nothing for three weeks. Seven lakh rupees still passed straight through a hedge that looked perfect on paper.

Why this matters

Week 11 reviewed equity decisions monthly. Week 12 updated bond valuations nightly. In derivatives a position can be correct in the morning and incorrect by lunchtime — and this topic shows why.

Following the hedge past where Week 10 stopped

Return to the hedge built in Week 10. Meera's fund is worth ₹5 crore with a beta of 1.2, so ₹5 crore $\times$ 1.2 gives a hedge value of ₹6 crore; one Nifty futures contract is worth about ₹6 lakh; so she shorts 100 contracts. At the time we said she had effectively switched off the market for a few weeks.

Three weeks pass and Meera trades not a single contract. But three things have quietly changed. Her **portfolio beta is no longer 1.2** — two large holdings have risen while three others declined, so the composition and sensitivity have shifted. The **market's mood has changed**, and in a nervous market her stocks may move more sharply relative to the index. And the **futures contract has moved closer to expiry**, so the basis has narrowed.

The fourth Monday

The market falls 4%. Meera's stocks lose ₹28 lakh. Her short futures position gains ₹21 lakh. **₹7 lakh** has passed straight through a hedge that looked perfect on paper.

Where did it go? Had beta stayed at 1.2 and both legs moved cleanly with the index, the portfolio would have lost ₹24 lakh and the futures gained ₹24 lakh — a perfect offset. The ₹28 lakh loss implies the portfolio actually moved 5.6%, so **beta had drifted to roughly 1.4**. The ₹21 lakh gain implies the futures leg moved about 3.5% against the index's 4% — the narrowing basis.

Nobody made a mistake and nothing failed. The hedge ratio that was correct on day one simply stopped being correct, and nobody was monitoring the change.

Why a straight line cannot describe a derivative

There is a second, mathematical reason derivatives need different tools. Recall Rahul's call option. The payoff stays completely **flat** until Zomato reaches the ₹270 strike, bends sharply upward at exactly that point, and rises thereafter.

Compare that shape with the best straight line a simple regression could draw through it. The line is inaccurate across most of the payoff, and worst of all *near the strike* — precisely the region that matters most to an options trader.

The bend is not a minor inconvenience. It is the essential nature of a derivative. You have already seen the sharp V of a straddle, the flat bottom of a strangle, and the ceiling created by a covered call. These structures contain corners, bends and cliffs — and even the smooth curves used for yield curves in Week 12 are not flexible enough.

WHERE THE ₹7 LAKH WENT

	If nothing had drifted	What actually happened	The difference
Portfolio move	$4\% \times 1.2 = 4.8\%$	5.6% — implying beta had drifted to 1.4	Beta drift
Portfolio result	–₹24 lakh	–₹28 lakh	–₹4 lakh
Futures leg move	4%, matching the index	About 3.5% — the basis had narrowed	Basis convergence
Futures result	+₹24 lakh	+₹21 lakh	–₹3 lakh
Net outcome	Perfect offset — nil	–₹7 lakh	Both legs drifted

Nobody made a mistake and nothing failed. The hedge ratio that was correct on day one simply stopped being correct, and nobody was watching.

Figure 1.1 The gap decomposed. Beta drift accounts for ₹4 lakh and basis convergence for ₹3 lakh.

KEY IDEA

A derivatives position decays even when nothing is traded. Every input that made it correct — beta, basis, volatility, moneyness — is moving continuously, which is why **a hedge must be monitored rather than set**.

EXAM POINTER

Decompose the gap rather than describing it: state what would have happened with no drift, then attribute the shortfall to beta drift and basis narrowing separately. For the mathematical point, explain that a straight line is least accurate exactly at the strike, where accuracy matters most.

QUICK RECAP · TOPIC 1

- Meera's hedge lost ₹7 lakh over three weeks in which she traded nothing.
- Beta drifted from 1.2 to about 1.4, and the basis narrowed as expiry approached.
- Neither drift was an error; both were unmonitored.
- Derivative payoffs bend sharply at the strike, where a straight line fits worst.
- Straddles, strangles and covered calls all contain corners a linear model cannot represent.

2 Decision Trees

PART A • WHY DERIVATIVES NEED DIFFERENT TOOLS • TOPIC 2 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Volatility is conditional. A model that says “one rule applies, except on event days” needs branches, not a straight line.

Why this matters

Trees are the first technique in the course that handles conditionality directly — and the reason they suit derivatives is that derivative behaviour is conditional almost everywhere.

The question a desk asks every morning

How volatile will the market be tomorrow? Note that this is *not* about direction — not whether the market rises or falls, but how large the movement will be. It is among the most valuable forecasts in derivatives, because it drives option prices, hedge sizes and margin requirements simultaneously.

A regression would answer with one straight-line formula applied to every day in history. But every experienced trader knows volatility is **conditional**: it behaves one way in a quiet market, differently when the union budget is three days away, and differently again when the market has already fallen sharply last week. A straight line cannot easily say “one rule applies, except on important event days”. A decision tree can.

Reading a volatility tree

The **India VIX** measures expected volatility over the next 30 days, published by the NSE as a real-time number derived from Nifty 50 option prices.

First question: is the India VIX above 18? If yes — is there a scheduled event, such as the budget or a results announcement, within the next five days? If yes as well, the model forecasts high volatility and the desk may increase its hedge. If the VIX is below 18 — has the Nifty moved more than 1.5% during the previous three days? A market that looks quiet but has just moved sharply may be less stable than one genuinely inactive.

Following the answers down the branches produces tomorrow’s forecast. Crucially, **the machine chose every question in the tree**.

Three properties worth knowing

Nobody wrote the questions. The machine tested thousands of possible divisions in historical data, across different features and threshold values, keeping those that best separated calm days from volatile ones. What looks like a trader’s rulebook was learned from data.

Trees capture interactions. An interaction means one factor matters only when another condition holds — an event within five days may mean one thing in a nervous market and something else in a calm one. Regression generally assumes each input contributes similarly in all situations; a tree does not.

Trees are readable. A tree can be printed, examined and discussed by a risk committee, and every forecast traced through a clear sequence of questions.

A VOLATILITY TREE • EVERY QUESTION CHOSEN BY THE MACHINE

IS INDIA VIX ABOVE 18?

The starting split, selected from thousands tested

IF YES • EVENT WITHIN 5 DAYS?

Budget, results announcement, policy decision

HIGH VOLATILITY FORECAST

The desk may increase its hedge

And down the other branch: if **VIX is below 18**, has the Nifty moved more than 1.5% in the previous three days? A market that looks quiet but has just moved sharply may be **less stable** than one that has been genuinely inactive.

Figure 2.1 A learned rulebook. The structure looks like trader intuition because it was fitted to the same reality.

REAL-WORLD EXAMPLE

What a tree can do on a derivatives desk: forecast volatility; estimate how much a large order may move the price before it is placed; classify the current market as calm, trending or stressed; and identify periods when an existing hedge is likely to become ineffective — exactly the problem nobody was monitoring in Topic 1.

COMMON PITFALL

Classifying market days is a *classification* problem, just like classifying borrowers in Week 12 — so it must be evaluated the same careful way. A regime model may report 90% accuracy simply because most days are quiet, while failing to identify the rare stormy ones. The most dangerous error is calling a highly volatile day calm: the equivalent of a missed default.

EXAM POINTER

Explain why volatility forecasting suits trees rather than regression — conditionality and interactions. State that the machine selects the questions by testing thousands of splits. Note the readability advantage, and that regime classification must be judged by a confusion matrix rather than overall accuracy.

QUICK RECAP • TOPIC 2

- The daily question is how large tomorrow's move will be, not its direction.
- Volatility is conditional, so a single straight-line rule cannot describe it.
- A tree asks a sequence of learned questions — India VIX above 18, event within five days, recent large moves.
- The machine chooses every split by testing thousands of possibilities in historical data.
- Trees capture interactions and remain readable enough for a risk committee to discuss.

3 From One Tree to a Forest

PART A • WHY DERIVATIVES NEED DIFFERENT TOOLS • TOPIC 3 OF 9 • SERVES LEARNING OBJECTIVE 2



IN ONE LINE

Sixty mediocre trees averaged together beat one carefully grown tree — for exactly the reason a portfolio beats a single stock.

Why this matters

This is the most satisfying connection in the module: the Markowitz principle from Week 5, applied to models rather than assets.

Why a single tree becomes greedy

Allowed to grow too many branches, a tree keeps dividing the data until it has a small rule for almost every unusual historical day — some of which were unusual only by chance. A very deep tree **memorises the past instead of learning a pattern that will work in future**. That is overfitting, named in Week 11.

The solution

Instead of one tree, build many. Picture the same problem two ways. One tree chasing every historical observation produces predictions that jump sharply up and down as it follows random noise. Now build **60 trees**, each trained on a random sample of the same data, and average their predictions. The result is smoother and tracks the underlying pattern more closely — not because any individual tree improved, but because **the errors of different trees partly cancelled**.

This collection is a **random forest**, and the logic should feel familiar. It is the portfolio principle from Week 5: several individually unreliable components combine into something more stable when their errors are not perfectly correlated. A random forest is a diversified portfolio of models.

Feature importance, and how it catches nonsense

Feature importance ranks inputs by how much they contributed to predictions. If the top inputs are the current India VIX, the scheduled event calendar and last week's market movement, a trader would find that reasonable — it matches how volatility is commonly understood.

If the model instead reports that the most important input is the *day of the month*, that should raise concern: the model has probably found a historical coincidence rather than a genuine relationship. Feature importance is not proof a model is correct, but it is a useful way to spot one that has learned something meaningless.

TWO WAYS TO BUILD FROM TREES

RANDOM FOREST**HOW TREES ARE BUILT**

Independently, each on a random sample

HOW PREDICTIONS COMBINE

Averaged across all trees

WHY IT WORKS

Errors of different trees partly cancel

ANALOGY

A diversified portfolio of models

GRADIENT BOOSTING**HOW TREES ARE BUILT**

Sequentially, each correcting the previous trees' mistakes

HOW PREDICTIONS COMBINE

Improved step by step, learning from earlier errors

WHY IT WORKS

Each tree targets what the others got wrong

NOTE

Extreme gradient boosting is a refined version, widely used in quantitative finance

*The random forest is **Markowitz applied to models**. Several individually unreliable components combine into something more stable when their errors are not perfectly correlated.*

Figure 3.1 Forests average independent trees; boosting builds them in sequence to fix each other.

KEY IDEA

Diversification works on *models* for the same reason it works on assets: independent errors partly cancel. That is why sixty mediocre trees can outperform one carefully grown one — and why correlation between models matters as much as correlation between stocks.

EXAM POINTER

Explain overfitting in a tree specifically — too many branches memorising unusual days. Distinguish forests from boosting by *how the trees are built*: independently and averaged, against sequentially and corrective. Then draw the Markowitz parallel explicitly; it is the connection the question is looking for.

QUICK RECAP · TOPIC 3

- A single deep tree memorises unusual historical days rather than learning a pattern.
- A random forest builds many trees on random samples and averages their predictions.
- It works because independent errors partly cancel — the Markowitz principle applied to models.
- Gradient boosting instead builds trees sequentially, each correcting the previous ones' mistakes.
- Feature importance ranks inputs and can expose a model that has learned a coincidence.

PART B • VOLATILITY AND SIMULATION

The number nobody can see

4 Volatility: The Input You Cannot Observe

PART B • VOLATILITY AND SIMULATION • TOPIC 4 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

Every option price in the world contains a forecast inside it — and Black-Scholes is used in reverse to get it out.

Why this matters

Week 9 deliberately simplified this. Volatility is the one pricing input that cannot be read off a screen, which makes it the most important quantity on a derivatives desk.

The input that is a forecast

When Rahul bought his call for a ₹12 premium, we said the premium depended on the spot price, the strike price and market volatility. The first two can be read off a screen. Volatility cannot.

Volatility is the market's opinion about how much the underlying may move between today and expiry — a forecast presented as an input. Which means every option price in the world contains a forecast inside it. The Week 9 option chain held eleven strike prices; those eleven prices also carried eleven implied volatilities.

Volatility clusters

Look at three years of daily Nifty returns and the movements occur in **clusters**. There are long stretches where the market barely moves, then stretches where almost every day brings a large movement. Calm follows calm; turbulence follows turbulence.

This leads to an important conclusion: **volatility is more forecastable than direction**. Nobody can consistently say whether the Nifty will rise or fall tomorrow. But we can often say something meaningful about whether tomorrow is likely to be calm or turbulent.

Traditional forecasting uses **GARCH** models — fitted using the maximum likelihood method from Week 12. You need not learn the mathematics, only what they do: they treat volatility as a quantity that changes over time, rising after large movements and gradually declining towards normal. Machine learning extends this by adding options market signals, trading volume, the event calendar, and the sentiment information from Week 11.

Black-Scholes as a translator

The **Black-Scholes model**, published in 1973, calculates an option price from five inputs: spot price, strike price, time to expiry, interest rate and volatility. Four of the five can be observed or calculated directly. Volatility is the only one that cannot.

In actual trading the option price is already visible on the exchange. So traders **run the model in reverse**: supply the observed price and the four known inputs, then solve for the volatility that makes the model price equal the market price. The result is **implied volatility**.

Black-Scholes is therefore used in practice as a *translator* — converting a price in rupees into a statement about how much the market expects the underlying to move.

ONE FIXED NUMBER, OR A DAILY FORECAST

ONE FIXED VOLATILITY VALUE**BEHAVIOUR**

The same number for the entire period

PROBLEM

Far too high in quiet periods, far too low in turbulent ones

VERDICT

Ignores the market it is describing

A DAILY UPDATED FORECAST**BEHAVIOUR**

Rises before and during stress, then falls gradually back towards normal

ADVANTAGE

Responds to current conditions

VERDICT

Reflects the market

Volatility is more forecastable than direction. Nobody can say whether the Nifty rises tomorrow. We can often say whether tomorrow is likely to be calm or turbulent.

Figure 4.1 Volatility clusters, so a fixed value is wrong nearly all the time.

KEY IDEA

Black-Scholes is famous as a pricing formula and used in practice as a **measuring instrument**. Nobody feeds it a volatility to get a price; they feed it a price to get the volatility the market is assuming.

EXAM POINTER

Name all five Black-Scholes inputs and identify volatility as the only unobservable one. Explain the reverse use and define implied volatility. State the clustering property and the conclusion it supports — volatility is more forecastable than direction — and name GARCH as the traditional approach.

QUICK RECAP • TOPIC 4

- Volatility is the market's forecast of future movement, presented as a pricing input.
- It cannot be observed, unlike the other four Black-Scholes inputs.
- Volatility clusters: calm follows calm and turbulence follows turbulence.
- That makes volatility more forecastable than direction.
- Traders run Black-Scholes in reverse to extract implied volatility from an observed price.

5 The Volatility Smile and Surface

PART B • VOLATILITY AND SIMULATION • TOPIC 5 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

The textbook says one volatility applies to every strike. The market disagrees, and the shape of that disagreement is itself information.

Why this matters

The smile is where the classical model visibly fails, and understanding why is what separates using a model from believing it.

The shape

A classical textbook model assumes one volatility applies to every strike. The real market does not behave that way. Take options expiring on the same date, put strike price along the horizontal axis and implied volatility up the vertical. The classical model predicts a flat line.

The actual market often shows **higher implied volatility at lower strikes**, declining near the current market price, then rising again at higher strikes. That curved pattern is the **volatility smile**.

The lower-strike side usually sits higher than the upper-strike side, which reflects investor behaviour: **people pay more for protection against a decline**, because crashes are faster and more severe than rallies.

Draw a smile for each expiry date and place them together and they form a three-dimensional **volatility surface**, with each strike's horizontal position closely related to moneyness.

Why the smile is not an error

It is the market *rejecting* the textbook assumption that returns follow a neat bell-shaped distribution. Real markets have **fat tails** — extreme movements occur more often than a simple bell curve predicts — and the smile is the price the market puts on that reality.

The volatility crush

Implied volatility may rise before the budget. Once the announcement is made and the uncertainty is resolved, it can fall sharply. That sudden decline is a **volatility crush**, and it produces one of the most painful outcomes in options trading.

Suppose a trader buys a straddle before an important event, expecting a large move without knowing the direction. The market may move *exactly as expected* — and the trader can still lose money, because the implied volatility paid for beforehand collapses once the uncertainty is resolved. The premiums fall as the volatility disappears. The trader was right about the movement and still lost.

WHAT A MACHINE DOES WITH THE SURFACE

Task	What it means	Why it matters
Fit it	Quote prices for strikes that did not trade during the day	The same problem as pricing silent bonds in Week 12, now in the options market
Check it for arbitrage	A badly fitted surface can imply prices that violate basic financial relationships	Traders could exploit the inconsistency at the desk's expense
Forecast how it changes	The surface rises, falls, twists and reshapes around events	Tomorrow's premiums depend on it, not on today's

The smile is not a market error. It is the market pricing fat tails — the fact that extreme moves happen more often than a bell curve predicts.

Figure 5.1 The volatility surface as a working object rather than a curiosity.

COMMON PITFALL

Assuming that being right about the direction and size of a move guarantees a profit on a volatility strategy. If the implied volatility paid for beforehand collapses after the event, the position can lose even when the underlying does exactly what was predicted.

EXAM POINTER

Describe the smile's shape and its asymmetry, and explain the asymmetry by investor demand for downside protection. State that the smile reflects fat tails rather than mispricing. Define the volatility crush and give the straddle example — right about the move, wrong on the premium paid.

QUICK RECAP • TOPIC 5

- The classical model predicts one volatility for all strikes; the market shows a curved smile.
- Lower strikes carry higher implied volatility, because investors pay more for downside protection.
- Smiles across several expiries form a three-dimensional volatility surface.
- The smile prices fat tails — extreme moves are more common than a bell curve predicts.
- A volatility crush can make a correctly predicted move still unprofitable.

6 Simulation and Stress Testing

PART B • VOLATILITY AND SIMULATION • TOPIC 6 OF 9 • SERVES LEARNING OBJECTIVE 3



IN ONE LINE

When no formula exists, imagine the future ten thousand times and count what happens.

Why this matters

Simulation is what makes complex derivative portfolios valuable-able at all. It also produces something richer than the confidence bands of Week 12 — an entire distribution.

When a formula runs out

Some questions have a formula. In Week 10 we calculated a futures price by multiplying the spot price by one plus the interest rate for the period, giving ₹251.25 — clean and exact. Most real derivative questions do not.

What is a structure worth that pays only if the Nifty stays between 23,000 and 25,000 for three months? What is the probability that Meera's hedged portfolio loses more than ₹50 lakh in any week of the next quarter? What happens to a portfolio of 200 option positions if the market falls 8% overnight and volatility doubles?

When no formula exists, we imagine the future thousands of times and count what happens.

The four steps

One: describe how the market may move during a single day, including the volatility discussed above. Two: use random numbers to extend that daily movement forward to the expiry date, creating one complete possible future. Three: calculate what the position would be worth at the end of that future. Four: repeat thousands of times, perhaps 10,000, and average the results.

For a call struck at 25,000: collect the final index value from every path, count how many finished above the strike, calculate the payoff for each, average all the payoffs, and discount that average back to the present.

The great strength is that the payoff calculation can represent almost anything — a single option, a straddle, a portfolio of 200 positions with different strikes and expiries, or a structured product with several conditions attached. As long as we can describe what the position pays under a given future, simulation can value it.

Richer than a confidence band

Week 12 required every model-based bond price to carry a band showing its uncertainty. A simulation does not produce one estimate with a range attached — it produces the **entire distribution** of possible outcomes. With 10,000 results you can ask anything: the average, the best outcomes, the worst, or the probability of crossing a particular loss level.

The limitation that must never be forgotten

A simulation is only as reliable as the assumptions used to create it. The computer does not independently know how markets behave — it generates outcomes according to the analyst's instructions.

Assume returns follow a neat bell curve and the 10,000 simulated futures will contain almost no extreme crashes, producing a reassuringly calm risk report that fails completely when a real crash arrives. This is exactly why fat tails and the volatility smile matter.

FIVE SCENARIOS • AND NOTE WHICH TWO INVOLVE NO MARKET FALL AT ALL

Scenario	Assumption	What the desk learns
Normal week	Today's volatility continues	Expected profit and routine margin requirement
Budget shock	Market moves 4% suddenly; volatility doubles overnight	Whether the hedge still works, and how much cash margin may demand
March 2020 replay	Market falls 10% over three days; volatility triples	The resulting loss, and whether there is enough cash to fund margin
Volatility crush	Market barely moves, but implied volatility falls 30%	How much a portfolio of long straddles may lose
Liquidity freeze	Positions cannot be closed for two trading days	The additional loss created by the inability to exit

Notice the last two. Neither requires a dramatic fall. A portfolio can be properly hedged against direction and still suffer badly — because volatility fell, or because it could not trade. A stress test asking only “what if the market falls 10%” would miss both.

Figure 6.1 Stress scenarios. The dangerous ones are not always the dramatic ones.

REAL-WORLD EXAMPLE

You simulate 10,000 futures for Meera's hedged portfolio. In 500 of them it loses more than ₹1 crore — an estimated **5% probability**. But one more question must be asked: *how severe are the losses in those 500 cases?* Two portfolios can share the same 5% threshold while the average loss beyond it is ₹1 crore 10 lakh in one and ₹3 crore in the other. Same probability, very different disaster.

EXAM POINTER

Give the four steps in order and state what simulation can value that formulas cannot. Emphasise that it produces a full distribution rather than a point estimate. The examinable caution is the assumption dependency — and an honest report gives both the probability of a serious loss and the severity beyond it.

QUICK RECAP • TOPIC 6

- When no formula exists, simulate thousands of possible futures and count the outcomes.
- Four steps: describe daily movement, extend it randomly, value the position, repeat and average.
- Simulation can value almost any payoff, however complex the structure.
- It produces an entire distribution, not a single estimate with a band.
- It is only as good as its assumptions — assume a bell curve and the report will miss every crash.

PART C • THE DERIVATIVES DESK

Where the theory meets the cash

7 The Ledger That Never Sleeps

PART C • THE DERIVATIVES DESK • TOPIC 7 OF 9 • SERVES LEARNING OBJECTIVE 4

**IN ONE LINE**

In equities a wrong decision reduces value on paper. In derivatives it demands cash the same evening.

Why this matters

Part C moves to the desk itself. The daily settlement introduced in Week 10 turns out to be the clock around which the entire operation runs.

Marking to market, at desk scale

Recall Rahul's daily settlement from Week 10. He holds one long Zomato futures contract at ₹262. Every evening the exchange revalues every position at the closing price and settles that day's profit or loss in cash. When the margin balance falls below the required level, a margin call arrives.

What Week 10 introduced as an evening process is, on a professional desk, the clock around which everything runs.

Why this changes the nature of a mistake

In the equity market a wrong decision reduces portfolio value on paper. **In derivatives, a wrong position requires an immediate cash payment, funded that evening.**

When the trader cannot provide the cash, the position may be closed at the worst possible time — even when the long-term view eventually proves correct. The trader is forced out simply because the margin account ran short.

Leverage magnifies more than profit and loss

Rahul controlled ₹2,62,000 of exposure with ₹40,000, roughly 6.5 times leverage, so a 5% move in the share produced about a 32% move in his deposit.

But leverage also magnifies **the cost of reacting slowly**. An equity desk that spots a problem a day late loses a day of value. A leveraged derivatives desk that spots it a day late may lose the entire position.

Which frames the question everything else in this module answers: **not what the position did to us last month, but what it will do to us tomorrow.**

THE SAME FOUR EVENINGS, AND WHAT THEY NOW MEAN

Evening	Settlement	Margin balance	What it means at desk scale
Start	Initial margin deposited	₹40,000	Capital committed before any position exists
Day 1	+₹5,000 credited	₹45,000	Profit arrives as cash, not as paper gain
Day 2	-₹12,000 deducted	₹33,000	Loss leaves as cash the same evening
Day 3	-₹5,000 deducted	₹28,000 — the broker calls	Solvency, not judgement, now decides whether the position survives

Scale this to an institutional desk holding several thousand contracts and the same movements involve **crores rather than thousands** — which is why anticipating margin calls is a machine learning problem in its own right.

Figure 7.1 Daily settlement. The question is never what happened last month.

KEY IDEA

Derivatives convert judgement risk into **liquidity risk**. A correct view held without enough cash to survive the daily settlements is indistinguishable, in outcome, from a wrong one.

EXAM POINTER

State the distinction precisely: an equity loss is a paper loss until sold, a futures loss is a cash loss the same evening. Explain that leverage magnifies the cost of reacting slowly, and conclude with the framing — the relevant question is forward-looking, not historical.

QUICK RECAP • TOPIC 7

- Derivatives are marked to market every evening, and losses are settled in cash.
- A wrong equity position loses value on paper; a wrong futures position demands cash tonight.
- A trader who cannot fund the call is closed out, even if the view later proves correct.
- Leverage magnifies the cost of reacting slowly as much as it magnifies profit and loss.
- The operative question is what the position will do tomorrow, not what it did last month.

8 The Hedge That Watches Itself

PART C • THE DERIVATIVES DESK • TOPIC 8 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

The Greeks measure each sensitivity separately — and all of them move continuously, which is why derivatives automated early.

Why this matters

This topic solves the problem that opened the week. It also introduces the vocabulary of every professional options desk.

Fixing Meera's hedge

Meera's hedge failed gradually because its central number, the hedge ratio, was fixed on day one and never examined again. The weakness is not in the formula — contracts equal portfolio value times beta, divided by the value of one contract. The weakness is **beta**, which as Week 11 established is a pulse rather than a permanent characteristic.

A machine can re-estimate beta daily and update the hedge ratio, staying much closer to true sensitivity, so the cumulative hedging error stays small instead of growing through the month. This is not a new technique — it is the continuously updated beta from Week 11, now connected to a real trading decision.

The Greeks

An option's price responds to several different factors, so traders measure each sensitivity separately. You do not need the formulas — only the question each one answers, and the fact that **each changes continuously as the market moves**.

A desk holding 200 positions must know its total delta, gamma, vega and theta at all times, and must trade when necessary to keep them within approved limits. No person can do this manually through a trading day — which is why derivatives trading was among the first areas of finance to become highly automated, long before “artificial intelligence” became a common term.

What modern machine learning adds

Faster valuation. A large portfolio may need repricing thousands of times a day, some requiring slow simulation. A desk can train a model to *imitate* the slow pricing model: it receives the same inputs, learns from the prices the simulation produces, and afterwards generates almost identical prices in a fraction of the time. It is not creating a new theory of option valuation — it is reproducing an expensive calculation more efficiently.

Smarter hedging. A textbook may say a hedge should be rebalanced continuously. In reality every trade costs money, so the desk must judge when the improvement in protection justifies the expense.

Anticipating margin calls. Suppose a model reports on Monday that, given the current volatility forecast, there is a one-in-four chance the desk will need an additional **₹18 crore** of margin by Thursday. That changes Monday's actions: the cash can be arranged in advance under normal conditions, rather than raised on Thursday afternoon in a falling market when many other desks need cash too.

THE FOUR GREEKS • FOUR QUESTIONS, ALL CHANGING CONTINUOUSLY

Greek	The question it answers	Connects to
Delta	How much does the option's value change when the underlying moves by ₹1?	The hedge ratio of the options market — closely related to duration in Week 8
Gamma	How quickly does delta itself change?	Why a hedge correct at 24,000 may already be wrong at 24,400
Vega	How does the option's value change when volatility changes?	The volatility forecasting of Topics 4 and 5
Theta	How much value does the option lose simply because one more day has passed?	The gradual loss of time value from Week 9

***The central lesson.** A derivatives hedge cannot be created once and then ignored. The portfolio changes, beta changes, option sensitivities change, volatility changes, margin requirements change.*

Figure 8.1 The Greeks. A desk with 200 positions must know all four at all times.

REAL-WORLD EXAMPLE

Futures arbitrage illustrates the same continuous monitoring. The theoretical futures price should equal spot grown at the risk-free rate — ₹250 becoming ₹251.25 after one month — and machines watch that gap across hundreds of contracts because differences are small, may last seconds, and often disappear before a person could finish the calculation. Two cautions: the theoretical price itself depends on estimated dividends and financing costs, so an apparent error may be a wrong assumption; and a visible gap is not a tradable profit once brokerage, taxes, spreads and market impact are counted.

EXAM POINTER

Name all four Greeks with the question each answers — that is the examinable content, not the formulas. Explain why derivatives automated early: every Greek changes continuously and a desk holds hundreds of positions. Give at least two of the three machine learning contributions.

QUICK RECAP • TOPIC 8

- Meera's hedge failed because the hedge ratio was fixed while beta drifted.
- Delta measures sensitivity to the underlying; gamma measures how fast delta changes.
- Vega measures sensitivity to volatility; theta measures the daily loss of time value.
- All four change continuously, which is why derivatives desks automated decades ago.
- Machine learning adds faster valuation, smarter rebalancing and anticipated margin calls.

9 Reinforcement Learning and Execution

PART C • THE DERIVATIVES DESK • TOPIC 9 OF 9 • SERVES LEARNING OBJECTIVE 4



IN ONE LINE

A forecast cannot answer how to sell ₹40 crore across a day. That needs a sequence of decisions — and a simulator the system can practise in.

Why this matters

The closing topic of the course. It is also where the most exaggerated claims about machine learning in finance are usually made, so it ends on what the technique genuinely does and does not do.

A different kind of question

Meera must sell ₹40 crore of Nifty futures during the trading day. Sell it all at once and her own order pushes the price against her before it completes — **market impact**, a real and measurable cost. Spread it too slowly and she stays exposed to market movements for hours — **market risk**.

Notice the kind of question this is. The models of Weeks 11 and 12 cannot answer it. A liquidity model may estimate how easily an asset can be traded at one moment — that is a forecast. Execution asks something different: *given that she must sell, in what sequence and at what speed should she trade so her own orders cause the least damage?* A forecast alone cannot answer that. She needs a **plan containing a sequence of decisions**.

What the machine discovered

Three strategies. Selling everything at once is fast but expensive — it exhausts available depth and pushes the price against her. Equal slices all day is more gradual, but trades just as heavily through the quiet, illiquid middle of the day as at the liquid open and close. Adjusting to market depth trades more when the market can absorb volume and less when depth is low.

Nobody programmed the third rule. The machine discovered it by experimenting across millions of simulated trading days. Market depth is generally high near the open, lower through the middle of the day, and rises again near the close.

Two features make this suit reinforcement learning. First, the decisions form a **sequence**: the amount traded now changes what remains to trade later. Second, the result is **delayed**: Meera may not know whether her morning decisions were good until the whole order is complete. A simple prediction model has neither characteristic.

Where it genuinely helps — and where it does not

The system usually learns inside a **simulator**, and knows only the kind of market that simulator represents. This deserves careful attention, because reinforcement learning attracts exaggerated claims. A system must practise a task millions of times, and no institution can let it lose real money millions of times to gain experience.

If the simulator contains no major crashes, the system never experiences one during training and may behave unpredictably when it meets its first real crisis. And real markets *respond* to a trader's actions — a large order changes available liquidity and moves the price. Many simplified simulators do not reproduce this, so a system trained against a simulator that never pushes back may be far too aggressive in a real market.

Its strongest practical use is **trade execution**: narrow, fast, repetitive decisions with a short feedback loop, where the effect of each individual decision is limited. That is a real and valuable application. Reinforcement learning is *not*, despite popular claims, a machine that learns to predict markets by playing against them.

THREE EXECUTION STRATEGIES • ONLY THE THIRD WAS LEARNED

Strategy	What it does	Result
Sell everything at once	The entire order in the first 15-minute interval	Fast, but expensive — it exhausts available depth and pushes the price against her
Equal slices all day	The same amount in every interval	More gradual, but trades as heavily through the illiquid middle of the day as at the open and close
Adjust to market depth	More when the market can absorb volume, less when depth is low	Follows the pattern of liquidity through the day — and nobody programmed this rule

***Exercise.** A policy trained on three calm years is deployed on a day when the market falls 6% and liquidity disappears. What should you expect? **Behaviour that cannot be confidently predicted** — the model has never seen this state. Every serious implementation needs hard risk limits and a manual stop.*

Figure 9.1 Execution strategies. The learned policy tracks liquidity because it was rewarded for doing so.

KEY IDEA

Reinforcement learning suits problems that are **sequential and delayed** — where each decision changes the next, and the verdict arrives only at the end. Forecasting is neither, which is why the two techniques are not substitutes.

EXAM POINTER

Explain why execution is not a forecasting problem — sequence and delayed feedback. Name the three strategies and note that only the third was learned. State both simulator limitations, and be precise about the honest claim: reinforcement learning is strong at execution, not at predicting markets.

QUICK RECAP • TOPIC 9

- Execution asks how to trade, not what will happen — a sequence of decisions, not a forecast.
- Selling at once costs market impact; selling too slowly leaves market risk.
- The machine learned to trade with available depth, a rule nobody programmed.
- It suits reinforcement learning because decisions are sequential and feedback is delayed.
- Policies learn inside simulators, so hard risk limits and a manual stop are non-negotiable.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain why linear models fail on derivatives and describe trees and forests	Topics 1, 2 and 3 • Figures 1.1, 2.1, 3.1
2	Explain implied volatility and interpret the smile and surface	Topics 4 and 5 • Figures 4.1, 5.1
3	Describe Monte Carlo simulation and design a stress test	Topic 6 • Figure 6.1
4	Explain daily settlement, the Greeks and the execution problem	Topics 7, 8 and 9 • Figures 7.1, 8.1, 9.1

The module in six sentences

A hedge decays silently: Meera lost ₹7 lakh over three weeks in which she traded nothing, because beta drifted from 1.2 to about 1.4 while the basis narrowed. Derivative payoffs bend sharply at the strike, where a straight line fits worst, so decision trees learn conditional rules instead — and random forests average many trees so their errors partly cancel, which is the Markowitz principle applied to models. Volatility is the one pricing input that cannot be observed, so traders run Black-Scholes in reverse to extract it, and the resulting smile is not an error but the market pricing fat tails. When no formula exists, simulation imagines thousands of futures and produces an entire distribution rather than a single estimate — though it is only as good as the assumptions written into it. Derivatives settle in cash every evening, so a wrong position demands money tonight and a trader who cannot fund the call is closed out even if the view later proves right. And the Greeks measure each sensitivity separately, all of them moving continuously, which is why derivatives automated decades before anyone used the phrase artificial intelligence.

HOW THE NINE TOPICS CONNECT • AND HOW THE COURSE CLOSES

PART A • THE PROBLEM AND THE TOOLS

- 1 • The decaying hedge
- 2 • Decision trees
- 3 • Forests

TAKEAWAY

Payoffs bend; lines do not

PART B • VOLATILITY AND SIMULATION

- 4 • Implied volatility
- 5 • The smile
- 6 • Simulation

TAKEAWAY

Measure what cannot be seen

PART C • THE DESK

- 7 • Daily settlement
- 8 • The Greeks
- 9 • Execution

TAKEAWAY

Positions decay; watch them

Closing the programme. Week 11 improved portfolio construction and risk estimation. Week 12 applied the same ideas to bonds — estimating missing prices, measuring credit risk, finding the drivers of the yield curve. Week 13 completed the journey with derivatives. The objective was never to master the mathematics behind every algorithm, but to understand the problems these techniques solve and why they matter. **Finance and data science are no longer separate disciplines.**

Figure M1 How the nine topics connect, and where the thirteen weeks have arrived.

Glossary and self-check



Term	Definition
Decision tree	A model learning a sequence of conditional questions, applying different rules in different situations.
Interaction	Where one factor matters only when another condition also holds.
Random forest	Many trees built on random samples, their predictions averaged for stability.
Gradient boosting	Trees built sequentially, each correcting the previous ones' mistakes.
Feature importance	A ranking of inputs by how much they contributed to predictions.
India VIX	The NSE's real-time measure of expected 30-day volatility, derived from Nifty 50 option prices.
Volatility clustering	The tendency of calm to follow calm and turbulence to follow turbulence.
GARCH	A family of models treating volatility as changing over time, fitted by maximum likelihood.
Implied volatility	The volatility forecast hidden inside an option's market price.
Black-Scholes	The 1973 option pricing model, used in practice in reverse as a translator from price to volatility.
Volatility smile	The curved pattern of implied volatility across strikes, higher on the low-strike side.
Volatility surface	The three-dimensional structure formed by smiles across several expiry dates.
Fat tails	Extreme market movements occurring more often than a bell-shaped distribution predicts.
Volatility crush	The sharp fall in implied volatility once an event resolves the uncertainty.
Monte Carlo simulation	Generating thousands of possible futures and averaging the resulting payoffs.
Stress testing	Simulating deliberately severe and unfavourable conditions.
Delta	The change in an option's value for a ₹1 move in the underlying.
Gamma	The rate at which delta itself changes.
Vega	The change in an option's value when volatility changes.
Theta	The value an option loses as one more day passes.
Market impact	The adverse price movement a large order causes before it completes.
Market depth	How much can be traded without causing a significant price movement.

Term	Definition
Execution policy	A learned sequence of trading decisions, rather than a single forecast.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 Meera traded nothing for three weeks and her hedge still lost ₹7 lakh. Account for it.

Beta drift and basis convergence. A ₹28 lakh portfolio loss on ₹5 crore implies a 5.6% move against a 4% index fall, so beta had drifted from 1.2 to about 1.4 — costing ₹4 lakh. The futures leg moved only about 3.5% as the basis narrowed towards expiry — costing ₹3 lakh. Neither was an error; neither was monitored.

Q2 Why does volatility forecasting suit a decision tree better than a regression?

Because volatility is conditional. It behaves one way in a quiet market, another before a budget, another after a sharp recent move. A regression assumes each input contributes similarly in all situations; a tree can say “this rule applies, except when an event falls within five days” — capturing interactions directly.

Q3 How is a random forest related to portfolio theory?

It is diversification applied to models. Sixty trees each trained on a random sample will make different errors, and averaging them cancels much of that error — exactly as combining imperfectly correlated assets reduces portfolio risk without proportionally reducing return.

Q4 Four of the five Black-Scholes inputs can be read off a screen. What happens with the fifth?

Volatility is solved for, in reverse. Since the option's market price is already visible, traders supply the price and the four known inputs and solve for the volatility that makes the model price match. The result is implied volatility, and Black-Scholes functions as a translator from rupees into an expected movement.

Q5 A trader buys a straddle before the budget. The market moves sharply, exactly as predicted. Why might the trade still lose?

Volatility crush. Implied volatility rises before an event and falls sharply once the uncertainty is resolved. The trader paid an elevated premium for volatility that then disappeared, so the option premiums fall even though the underlying did what was expected.

Q6 A simulation shows a 5% probability of losing more than ₹1 crore. What else must the report say, and why?

How severe the losses are beyond that threshold. Two portfolios can share the same 5% probability while the average loss beyond it is ₹1.1 crore in one and ₹3 crore in the other. Probability alone describes the frequency of a disaster, not its size.

Q7 Why is trade execution a reinforcement learning problem rather than a forecasting one?

Because the decisions are sequential and the feedback is delayed. How much is sold now changes what remains to sell later, and the quality of the morning's decisions is unknown until the whole order completes. A forecast produces one number; execution needs a policy.

Q8 An execution policy trained on three calm years meets a 6% fall with vanishing liquidity. What should the desk have in place?

Hard risk limits that override the policy, and a manual stop. The system learned inside a simulator containing no such day, so its behaviour cannot be confidently predicted. Every serious implementation requires a human able to halt it immediately.